MILITARY SPECIFICATION SHEET

TRANSFORMER, POWER, STEP-DOWN TF5R032Z

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the transformer described herein shall consist of this document and the latest issue of Specification MIL-T-27.

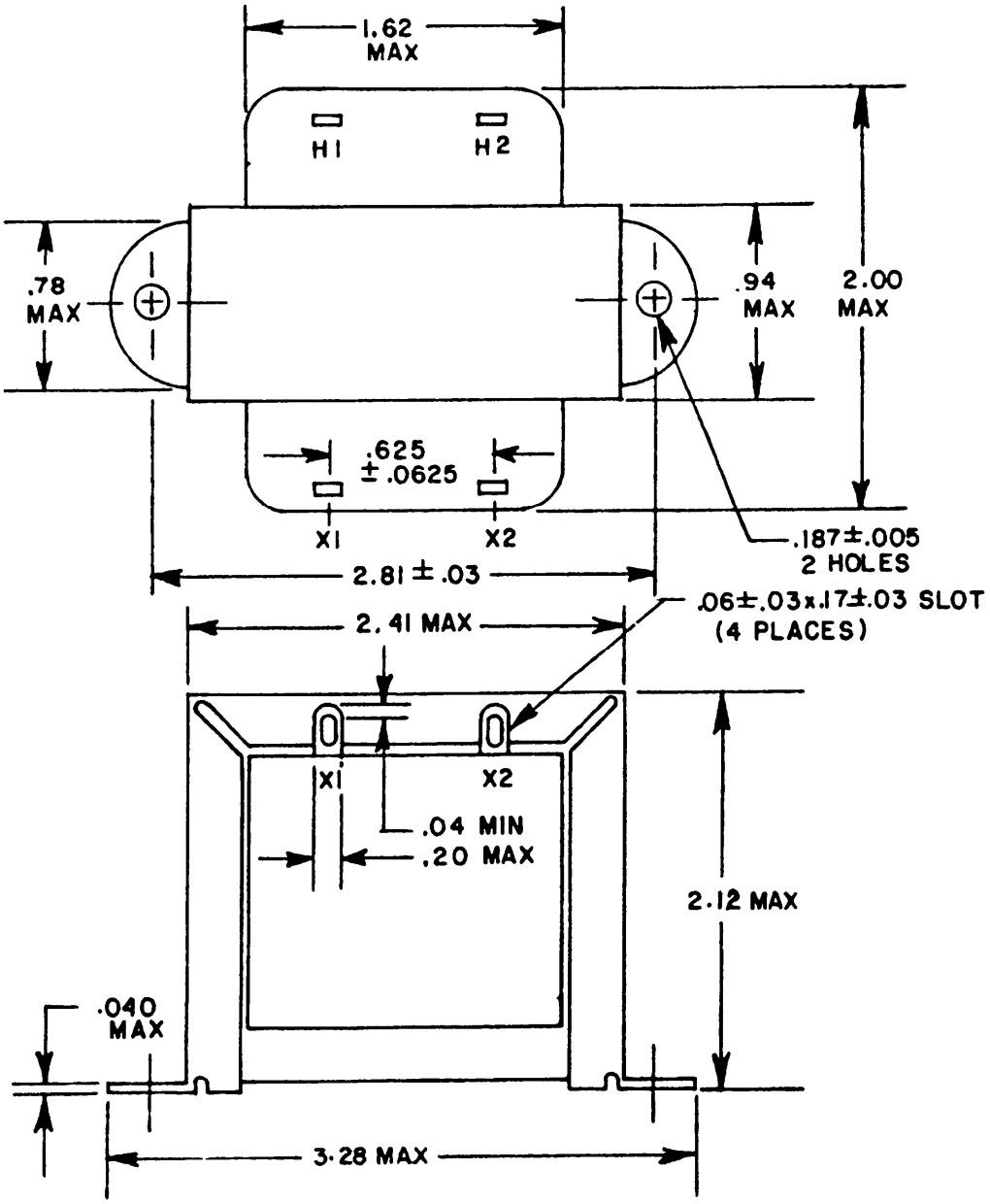
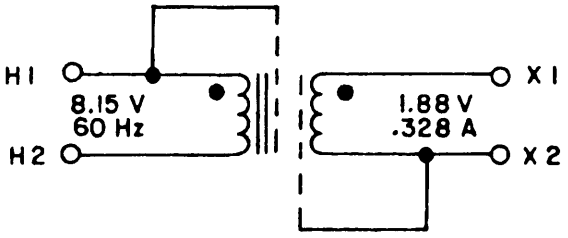


FIGURE 1. Dimensions and configuration.

FSC 5950

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WV - 857 V MAX
ALTITUDE - 10,000 FT MAX

CIRCUIT DIAGRAM AND MARKING.

INCHES	MM
.030	.76
.040	1.02
.060	1.52
.0625	1.588
.170	4.32
.200	5.08
.625	15.88
.780	19.81
.940	23.88
1.62	41.1
2.00	50.8
2.12	53.8
2.41	61.2
2.81	71.4
3.28	83.3

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. Marking shall be on the top of the case.

FIGURE 1. Dimensions and configuration - Continued.

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REQUIREMENT: (When numbers in parentheses, i.e., (H1-H2) are used, they indicate the winding and the extreme terminals of the winding.)

Electrical ratings:

Primary voltage: (H1-H2): 8.15 Vrms, 60 hertz.

Primary current: (H1-H2): .082 ampere rms.

Volt ampere: .67 volt ampere.

Secondary (X1-X2): 1.88 volts, .328 ampere rms.

Working voltage: 857 volts, maximum.

Design and construction:

Dimensions and configuration: See figure 1.

Duty cycle: Continuous.

Case: Encapsulated.

Material: Epoxy.

Terminals: Solder type.

Material: Brass, tinned or cadmium plated.

Height: 0.25 inch, minimum.

Weight: 16.7 ounces, maximum.

Operating temperature range: +5° to +105°C.

Terminal strength: MIL-STD-202, method 211, test condition A, 2.0 pounds.

Dielectric withstanding voltage (at sea level): 2,640 volts rms.

Electrical characteristics:

No load: With 8.2 volts, 60 hertz across (H1-H2), current in (H1-H2) shall not exceed 320 milliamperes and the power in (H1-H2) shall not exceed 2.5 watts.

Voltage across (X1-X2): 2.05 volts ±1 percent.

Rated load: With 8.2 volts, 60 hertz across (H1-H2) and .328 ampere resistive load in (X1-X2), the voltage across (X1-X2) shall be 1.88 volts ±5 percent.

DC resistance:

(H1-H2): 3.42 ohms ±20 percent.

(X1-X2): 0.31 ohm ±20 percent.

Electrostatic shielding: Static shields shall consist of 1 turn of 2 mil copper foil each.

Polarity: Additive, with terminals X1 and H2 connected together.

Temperature rise: 40°C maximum with rated voltage and current at an ambient temperature 65°C.

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Shock (high impact): MIL-STD-202, method 207.

Marking location: See figure 1.

Part number: M27/293-01.

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - MI
Navy - SH, OS
Air Force - 11, 17, 99
DLA - ES

User activities:
Army - AR, WC
Navy - MC, AS
Air Force - 19

Preparing activity:
Army - ER

Agent:
DLA - ES

(Project 5950-0568-10)

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DOCUMENT IDENTIFIER *(Number)* AND TITLE
TF5R03ZZ MIL-T-27/293 TRANSFORMER, POWER, STEP-DOWN

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

2. REMARKS

SUBMITTED BY *(Printed or typed name and address — Optional)*

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DD FORM 1 OCT 76 1426 EDITION OF 1 JAN 72 WILL BE USED UNTIL EXHAUSTED.

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